



37000: Marketing Strategy

New Products and Product Line Decisions

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Outline

- New Products
 - How do you come up with them?
 - How do you forecast demand for them?
- Product Line Decisions
 - When do you want to extend your product line and why?



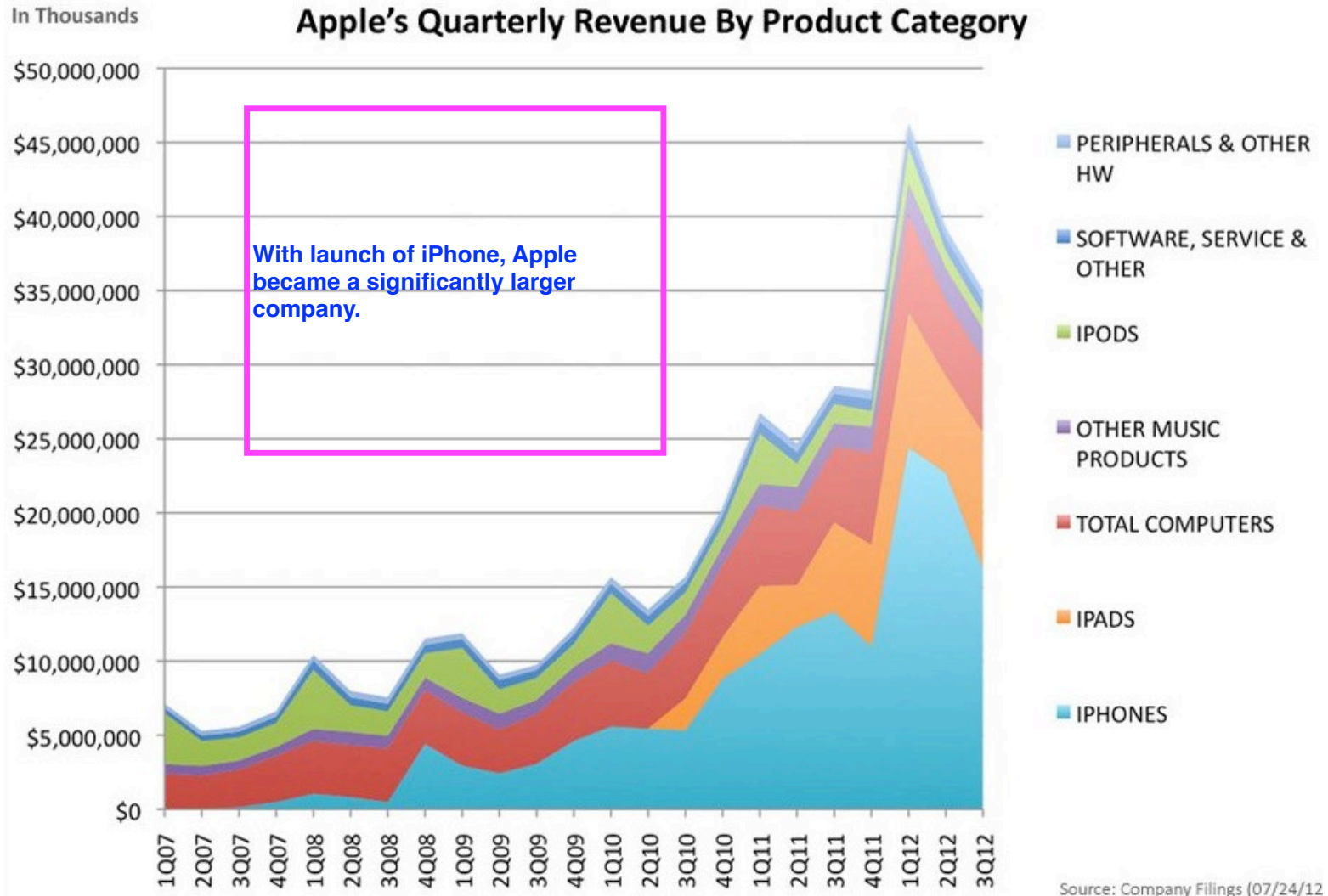
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New Products and the Product Lifecycle

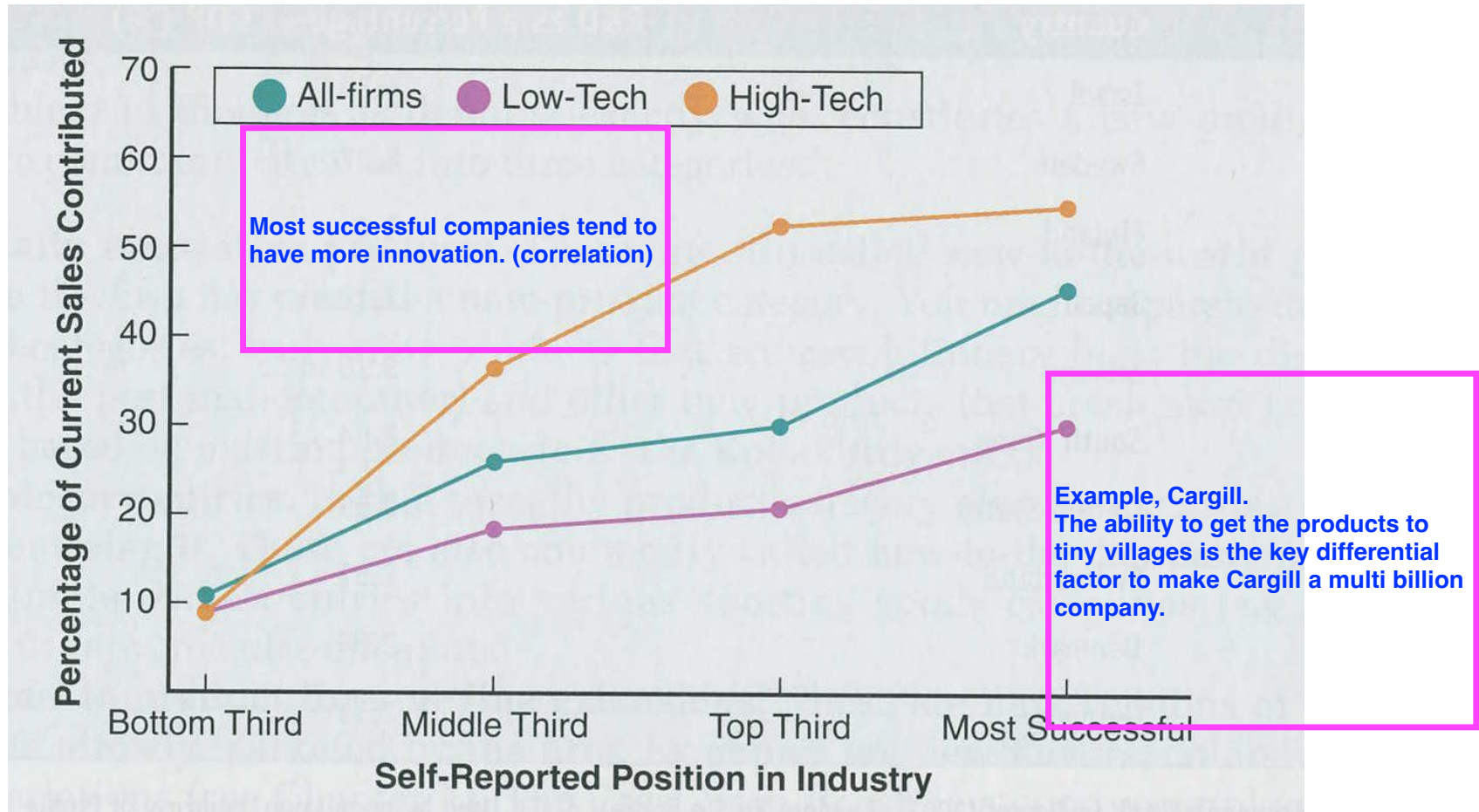
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The Value of New Products

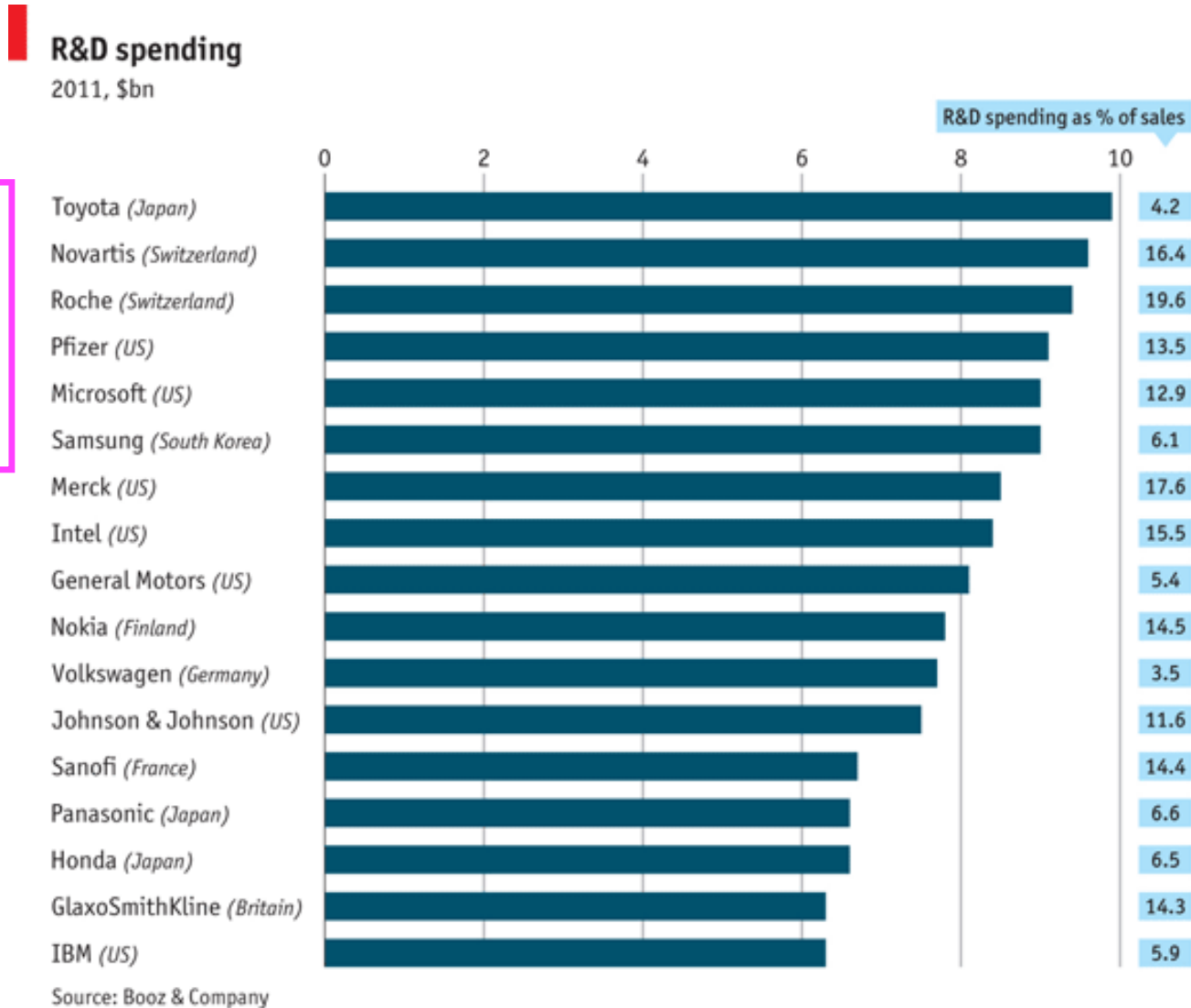


The Value of New Products

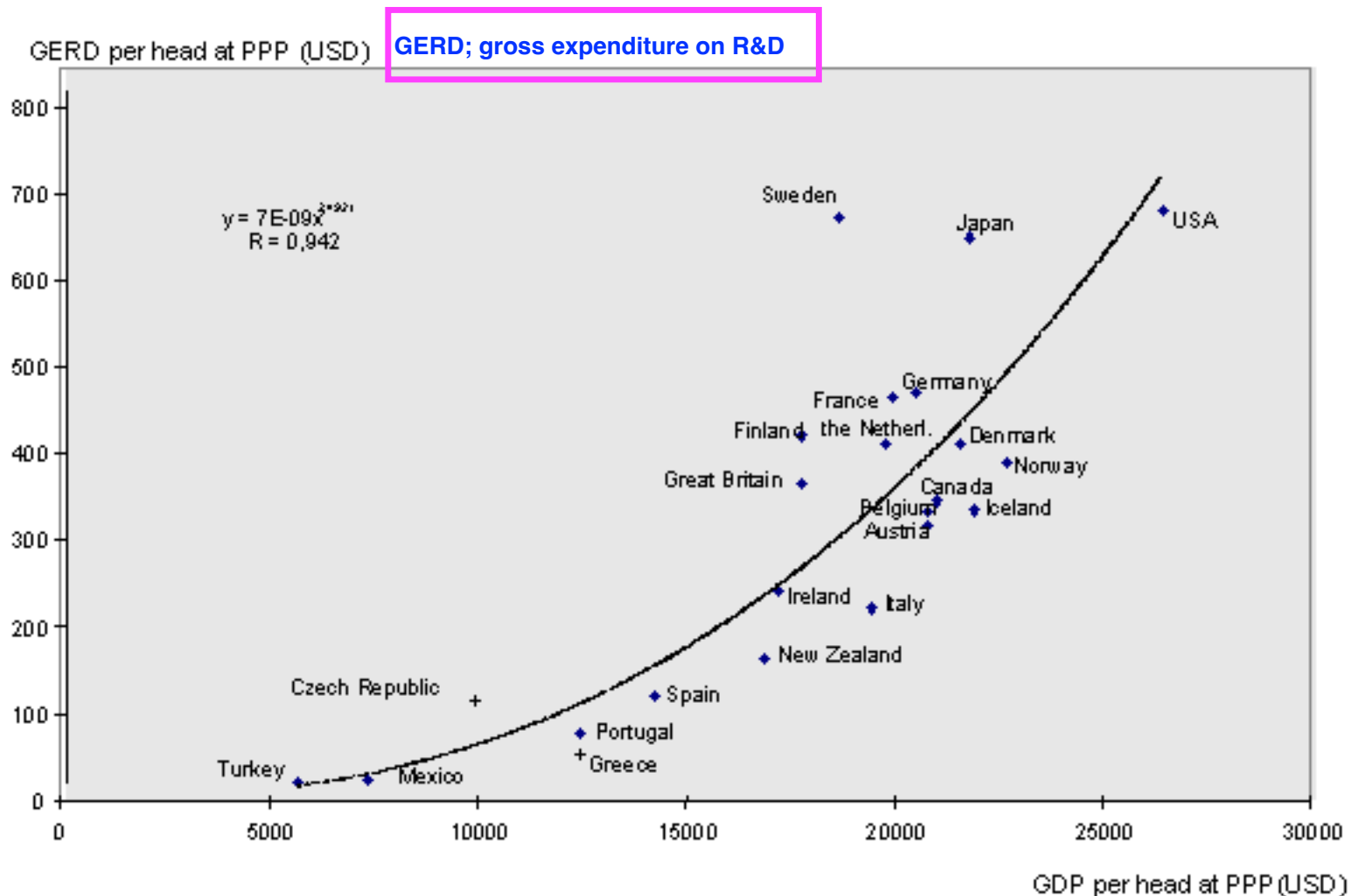


Sales Share of New-ish Products (<5 years)

The Value of New Products



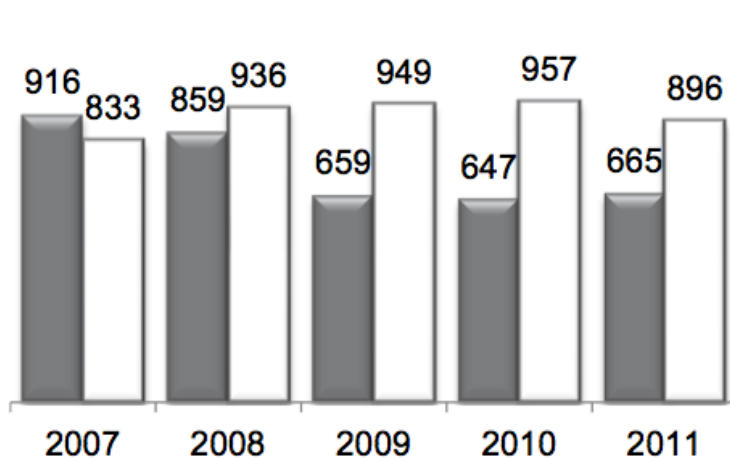
Pharma needs
to do new
products,
otherwise they
are dead



It is taken as axiomatic that innovative activity has been the single, most important component of long-term economic growth (OECD)

Failure of New Brands

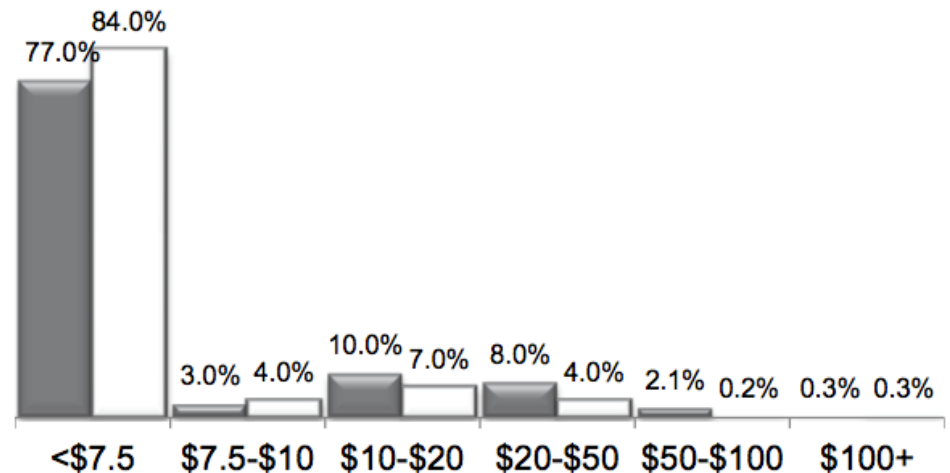
Number of CPG New Product Introductions*
Brand Level



■ Total Food & Beverage □ Total Non-Foods

Source: SymphonyIRI New Product Profiler™
*Across Food, Drug & Mass Channels (excluding Walmart)

2011 Proportion of CPG New Products
by Year-One Sales* (\$ Millions)



■ Total Food & Beverage □ Total Non-Foods

Source: SymphonyIRI New Product Profiler™
*Across Food, Drug & Mass Channels (excluding Walmart)

Why are many CPG new products failing? less than 7.5m in first year is really bad.
- possibly part of a poliferation strategy to fill the shelves.

Introduction – New Products

- How do you come up with new products?
 - At it's heart, innovation
- How do you forecast demand for them?
 - Building capacity, supply chain, distribution
- Lead to: Product Lifecycle

New Products

- Types of New Products

New Products

- Types of New Products
- Extensions:
 - Brand extension into new category or slight modifications of existing products
 - “Relatively” safe, easier to forecast demand, lower reward
- More radical variations on products:
 - New but relatable category or drastic innovation in existing category
 - Riskier but can still forecast
 - Potentially huge payoff
- Totally out there.
 - High risk but huge leaps towards a more ideal customer experience

What Matters Is....

- How do you come up with the idea in the first place?
 - Need: deep understanding of customer needs
- How do you measure demand / forecast the lifecycle?
 - Conjoint or Observational Demand Analysis!
 - Can be combined with diffusion models (I think Pradeep showed you a little bit of this)

New Product Creation



Pepsi-Co owns both brands. What would show up in cross reference data that might make this a good idea?

Taco Bell was rolling out another idea, the 4th meal (meal after drinks night). They identified high WTP from such drinking customers



Tide pods; marketer realise that there are people who regularly buy small pack size at higher price per volume.

Descriptor variable; people do not want to buy large bottles for a variety of reasons.

New Product Creation: Taco Bell

- 1) Mandate / Resources
- 2) Brainstorming session
- 3) Concept Development
 - Concept testing with consumers
- 4) Create product
 - Taste tests
- 5) Iteration
- 6) Production / Engineering challenges
- 7) Test markets
- 8) Full product line roll out
- 9) Future extensions

Sample Concept Test

Automated e-mail Responder (AeR)

The electronic age brings new headaches for the busy executives—junk e-mails. Now a group of acclaimed computer scientists have used the latest technology to develop a software which saves time of busy executives by automatically responding to e-mails.

What is AeR and How does it work?

AeR is an expert system based on pattern recognition technology. After you install this software, it works in three phases: learning, testing, and actual operation. In the learning phase, you respond to e-mails as usual and AeR “learns” from your way of responding. In the testing phase, AeR responds to e-mails and asks you to approve or disapprove the way it responded. AeR then adjusts its response style. Once you are satisfied, AeR is ready for actual operation where it automatically responds to most of your e-mails. If unsure about an e-mail, the software leaves it for you to respond.

How attractive do you find the idea behind this software?

- ☐ Very Attractive
- ☐ Somewhat attractive
- ☐ Slightly attractive
- ☐ Not attractive at all

If this software were available for a reasonable price, how likely are you to buy it?

- ☐ I will definitely buy
- ☐ I will probably buy
- ☐ I might or might not buy
- ☐ I will probably not buy
- ☐ I will definitely not buy

New Product Creation



Unholy Abomination Called 'Doritos Loaded' Tested At D.C.-
Area 7-Elevens

New Product Creation: Tide Pods

- 1) Start with a problem:
 - “Apartment dwellers hated lugging a detergent bottle downstairs”
- 2) Make the prototype (outsource parts)
- 3) Test in “lab”
 - Product itself and purchase environment
 - Different version of the product?
- 4) Distribution / Promotion

New Product Failure



Marketers found that most people who bought NETBOOK is women, and they tried to create the DELLA for women.

Bad failure! Why? Customer needs are email and work. And there is no difference in terms of this particular need between men and women.

- "Find recipes online, store and organize them, and watch cooking videos". You can "use your mini to track calories, carbs, and protein with ease".
- "Once you get beyond how cute they are, you'll find that netbooks can do a lot more than check your e-mail"

New Product Failure

- Which would you prefer?
 - A Soft, buttery homestyle bun. Fresh, crisp, lettuce. A juicy, quarter-pound patty of 100% pure domestic beef. Sweet, slivered, Spanish onions. A new “secret sauce” for grown-ups.
 - An ungodly chemical stew of robotically destroyed pig-parts, smoke-flavored, dyed with fake grill marks, and then, absurdly, molded with “rib” shapes on top

What is the need that Mc serves for the adults?

New Product Failure



INSIDE THE MIND OF THE CONSUMER



I DON'T SEE OUR
BRAND OF PICKLE
RELISH ANYWHERE.

INCONCEIVABLE,
THIS IS A "BRAND
LOYALIST."



TOM
FISH
BURN

What is a Brand?

- We talked about providing *trust*.
- But how? What exactly is a brand?
- **Unique Benefit.**
 - To meet customer segment *needs*.
 - Over the Next Best Alternative.
 - Consistent through time.

What makes a brand a brand?

-in the “needs” space. that your coy
consistently and uniquely deliver
on top and above the NB
alternative.

DCE

Desired Consumer/Customer Experience

**The ultimate benefit desired by the consumer/
customer that is perceived as almost unachievable**

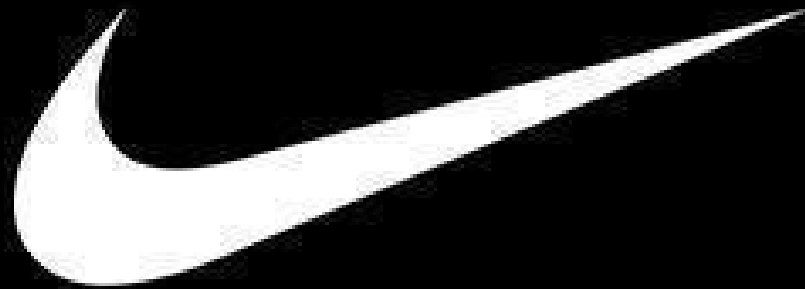


New Product Creation Goal

- Advance towards DCE.
- In a way that *maintains or advances* brand promise.
- **Unique Benefit.**
 - To meet customer segment *needs*.
 - Over the Next Best Alternative.
 - Consistent through time.

Cheapest airline with the most connectivity.
Getting people to go anywhere as cheap as possible.
- new product innovation for SPIRIT should be to make even cheaper flights.







More “Radical” New Products

- 1) What is the DCE in a category?
- 2) How Can I Move Market Towards that DCE?
 - How can i stretch the new product to the furthest point of the DCE base on the current (or soon) technology
- 3) Out of Category Extensions:
 - How does my brand promise *conceptually* help advance target category towards DCE?
 - How does my technical expertise allow me to do it?

VC: New Products as Real Options?

- 1) Need to start with a sense of consumer needs/DCE.
- 2) Spray & Pray.
 - Try many options across the spectrum of attributes people might like to meet their needs.
 - Tend to be startups with no real “brand”
 - Almost like a conjoint but in real life.
- 3) Risk in execution, value of particular attributes, etc.

akin to venture capitalist, where many new products may actually fail. Think of regression formula;

$$y = x\text{Beta} + e$$

be clear and strong in the xBeta aspect, which we can study and control. but acknowledge the e (market risk) that the product may fail.

Luck or Skill?

- 1) More radical new products inherently carry some risk.
 - Your analysis can be good, but your product still fails.
 - Your analysis can be not the best, but your product still succeeds by luck.
- 2) Good analysis will give you a better chance of success and a better direction of where to go next if you fail.

The Math: Launch new product?

- 1) Trivially easy given our framework.
 - Estimate consumer utility- conjoint, etc.
 - Set your 4 P's optimally given demand estimation.
 - In particular, price. But really any attribute that is flexible.
 - Given those 4 P's, forecast demand.
 - Compute total discounted profit over lifecycle
- If Total Expected Variable Profit > Fixed/Sunk Cost of Entering market → Launch. Otherwise, don't.
 - $Expected[\text{Variable Profit}] = \text{Prob}(\text{Success})[\text{Total Discounted Variable Profit}] + (1 - \text{Prob}(\text{Success})) * 0 < \text{Fixed Cost}$

Summary:

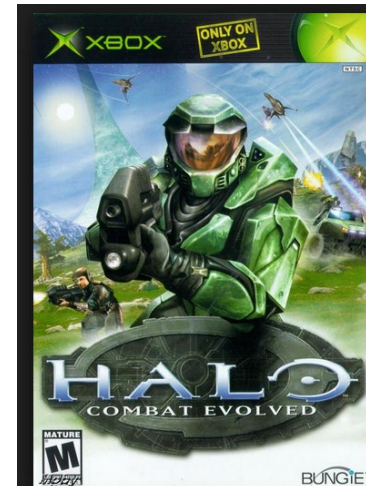
- New product development is challenging:
 - Deep understanding of consumer needs and preferences + creative genius in big leaps forward in ways to meet those needs.
 - Still a high chance of “failure”
- For less radical products:
 - Iteration between product design and customer input.
 - Look in your segmentation for unmet needs.
 - Always think in terms of consumer preferences (e.g. conjoint) and the competition space

New Products - Summary

- New product development:



- Consumer adoption decisions / forecasting





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Product Line Decisions

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Product Line Decisions Outline

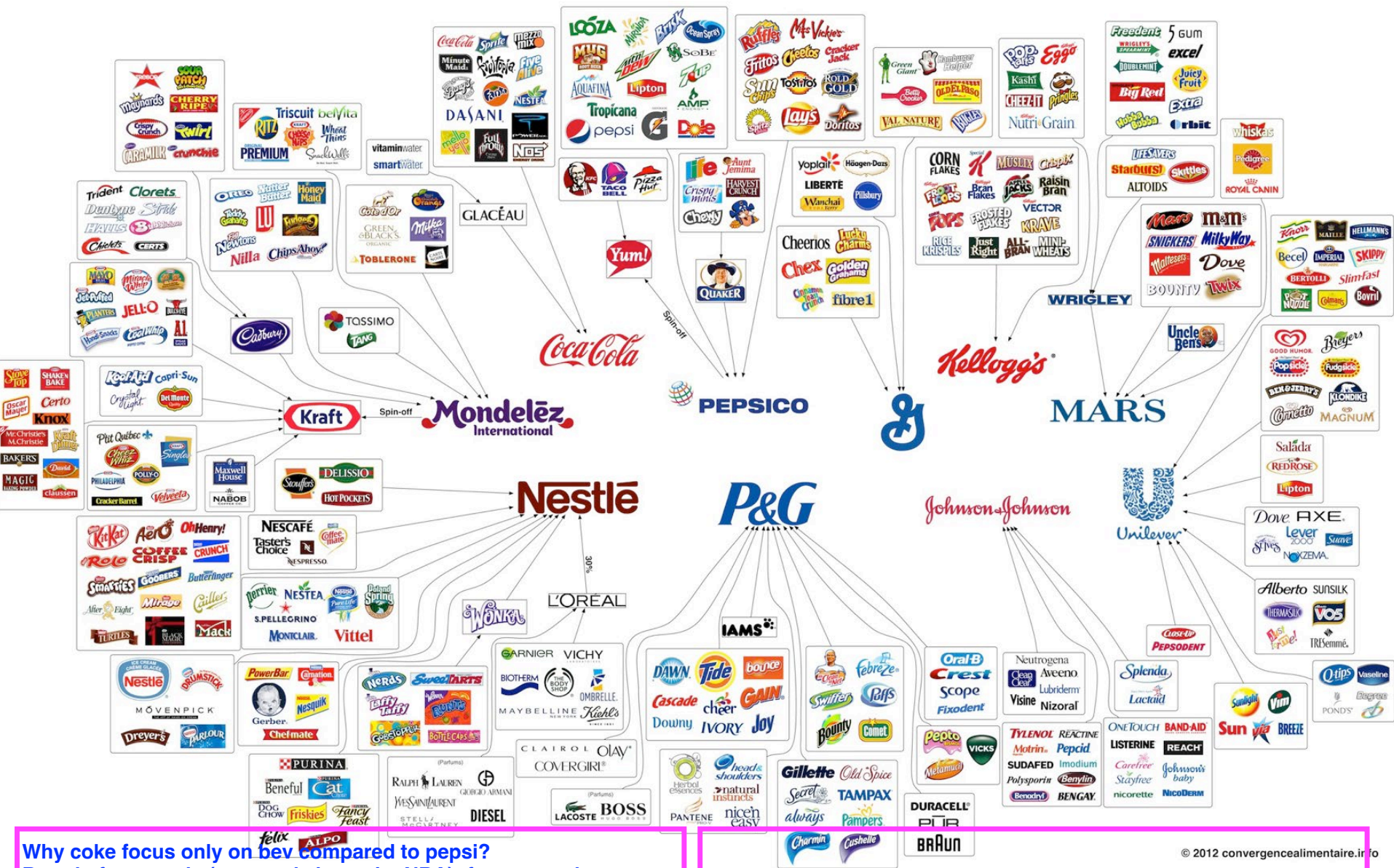
- Moving from a single product to two products and the tradeoffs
 - Behavioral components
- Deciding when product line extensions make sense vs. totally new products vs. not entering a market
- Where does Brand fit in?
- Behavioral Components

Do you Want to Have a Product Line?

- Question 1:
 - Do you want to introduce a product that is “unrelated” to your first-best target market product?
- Question 2:
 - Do you want to introduce a (second) product that is “similar” to your current product?
- Question 3
 - What about complementary products?
- Which is which?

Unrelated Product

- Why do we see some companies with large, seemingly unrelated product lines
- And others with focused, short product lines.



Why coke focus only on bev compared to pepsi?
 Brand of coca cola (over and above the NBA); focus on advert, brand, purported taste
 pepsi wins in blind taste because its sweeter (performs better in small batch taste)
 Pepsi is a distribution company (performs that better than NBA)

Why does Nestle have cosmetics (LOreal?) They have brand trust, they employ more scientist than anyone else, they have Swiss heritage.

Unrelated Products

- Synergies in:
 - Distribution
 - Manufacturing
 - Innovation
 - Branding
- Does it fit with our corporate culture?
- Is there marketing here?
 - Economies of scope
 - Operations
 - Inside the firm dynamics (OB)

Partial Substitutes

- More interesting from a marketing perspective:
 - When should you introduce a product that is similar to your existing product line?
- Considerations
 - New customers vs. Cannibalization (constrained optimization)
 - Brand dilution.
 - Behavioral Constraints and Considerations

Related products: Motivation



“Only too often does the sight of third-class passengers traveling in open or poorly sprung carriages, and always badly seated, raise an outcry against the barbarity of the railway companies.

It wouldn't cost much, people say, to put down a few yards of leather and a few pounds of horsehair.”

Motivation



“What the company must do is to prevent the passengers who can pay the second-class fare from traveling third class;

It hits the poor, not because it wants to hurt them, but to frighten the rich.

Having refused the poor what is necessary, they give the rich what is superfluous.”

Self-Selection

- How does consumer self-selection affect decisions about whether to create a second product or not.
- Fundamental framework:
 - Constrained Optimization

The Tradeoff

- Attract:
 - Consumer that wouldn't have purchased now purchase
 - Those that aren't in the category at all
 - Stealing business from your competitors
- Lose:
 - People that are purchasing your current product (which might be higher margin)
- Choosing product quality appropriately can mitigate this loss

fun note; airline.

when you book, you want cheap. when you travel you want comfort.
your future self (travel) hates your current self.

Motivation

3 May 2013

HUFFPOST BUSINESS

UNITED KINGDOM 

Third Class Rail Tickets: Standing Room Only Or Premium Economy Fares Could Be Introduced

“Getting a seat can be a struggle for some passengers as overcrowding on the railways continues to be a big problem – nationally less than three in four passengers (69 per cent) tell us that they are satisfied with the availability of seats or space to stand.”

Nissan & Infinity from Price Discrimination

- Should you introduce the Nissan?
- Depends on:
 - Marginal Costs
 - Distribution of consumer preferences
 - How many low type guys are out there.
- Choose:
 - Qualities / Prices of the products
 - Can derive the number of people that substitute
- You might want to lower the quality of your worse product so that you don't cannibalize your high margin product

Audi made an A1 to target the low WTP customer. how to prevent cannibalisation? -> intentionally create A1 to be small, so that the higher WTP customers will choose the larger cars.

Example: Should Apple Introduce a Cheap iPhone?

Apple to sell 75 million low-cost iPhones next year, says analyst

The much-rumored low-cost iPhone would cannibalize sales from the more expensive model but help Apple grab a healthy chunk of the low-end smartphone market.



by Lance Whitney | April 16, 2013 5:39 AM PDT



(Credit: CNET)

Example: Should Apple Introduce a Cheap iPhone?

In an investors note released today, Munster said he expects Apple to sell a \$300 non-subsidized iPhone starting in September. Such a device is likely to trigger a 30 percent cannibalization rate, which means that for every three low-cost iPhones sold, one full-price iPhone is cannibalized.

As such, Apple's share of the high-end smartphone market may dip to 37 percent next year from 43 percent last year. But its share of the low-end (under \$400) market will rise to 11 percent in 2014 from nothing in 2012.

The iPhone 5c

Despite Record Sales Quarter, Apple's iPhone 5c Likely Wasn't Key To Growth

Posted Jan 27, 2014 by [Darrell Etherington \(@drizzled\)](#)

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This suspicion that the iPhone 5c is not selling quite as well as Apple might have expected

it could indicate that Apple needs to really deliver on the idea of a cheaper iPhone – the iPhone 5c was anticipated to be that, but in the end it was the same price as older devices bumped down the line in previous quarters.

The \$17 Chianti vs. the \$72 Brunello



Both wines came from Tuscany.
in blind taste test, 80% of people chose the Chianti.
High WTP customers will buy the Brunello.

** most people who drink wine knows very little about wine. Brunello is specially for pairing (esp with truffles and some meats).

important; do not make a bad Chianti just to avoid cannibalisation.

Competition

- What if there is a competitor in the new market?
 - Positive: More consumers purchase my product
 - Negative: You have to lower price to attract any customers
 - This increases the cannibalization problem

avoid lowering price, because it may result in race to the bottom.
recall CS class, usually the weaker player will initiate price war.

Behavioral Components

- Other reasons not to extend your product line?
 - Choice overload
- Other reasons to extend your product line?
 - Context effects
- As usual, be careful in figuring out when and where these things apply.

Choice Overload



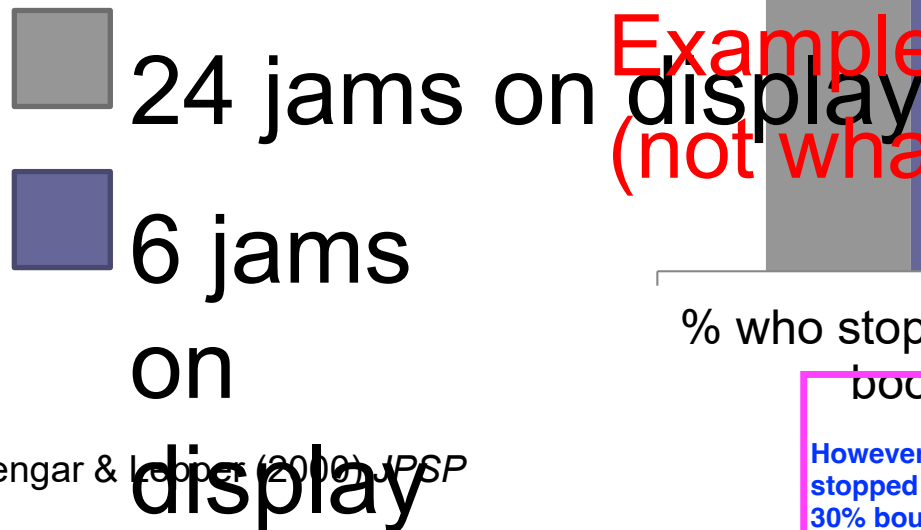
P&G: Real World

- Since so many of the 30,000 items in a typical supermarket offer small differences from each other, shoppers face numbing selection.
 - “There’s this vast array of products,” says Shelby Reyes, a 58-year-old Cleveland bookkeeper. “A lot of times I end up getting what’s on sale...or I say the hell with it.” Concedes P&G’s president, Durk I. Jager: “It’s mind-boggling how difficult we’ve made it for them over the years.”
- One of the biggest cuts came in its U.S. hair-care business. Minor brands such as Prell and Ivory shampoo were cut way back. Even the mighty Head & Shoulders line was pared in half, to 15 variations.
- The moves met some resistance from P&G’s brand honchos, who thought, “Oh my God, we’re going to lose sales because we’re going to have fewer items,”
- But simplification quickly paid off: Sales per item in hair care more than doubled

Grocery Store Example



Information Search: “Choice Overload”

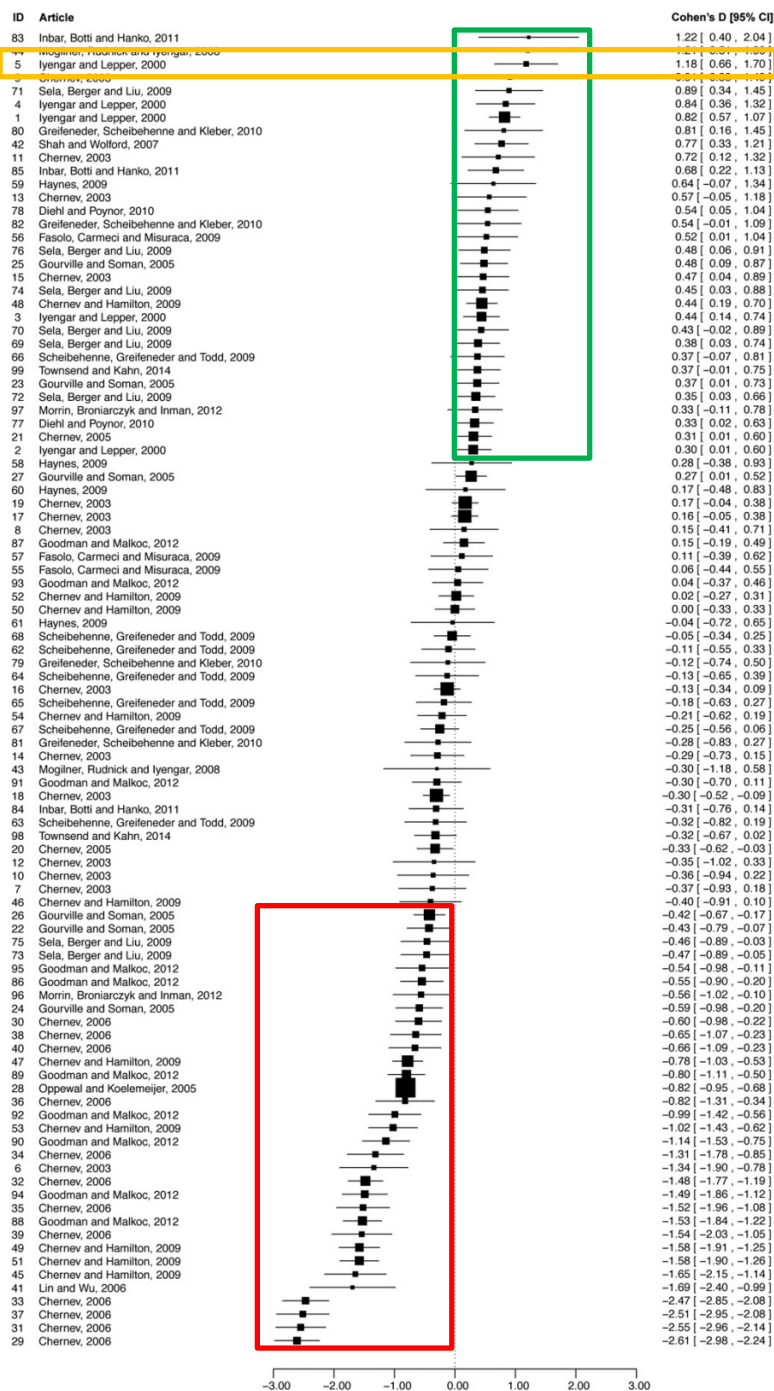


What does this tell us about real decision-makers?

Example of what CAN happen (not what always does happen)

% who stopped at the booth

However, fewer people bought when there is 24 jams (more people stopped to try something though). Only 3% bought, compared to 30% bought when there is only 6 jams on display. [missing graph on right]



Research: “Choice Overload”

What would you conclude?

How should we approach choosing the number of offerings?

jams study (highlighted in green) shows the positive correlation of choice overload. however, CO study are subjective and highly idiosyncratic.

Trader Joe's

Trader Joes, Where Less Is More

■ DENISE LEE YOHN, BRAND AS BUSINESS BITES |   | MAY 31, 2011, 8:22 AM |  1,565 | 

 Recommend { 21 }  Share { 12 }  Tweet { 23 }  +1 { 0 }  EMAIL  MORE

- Trader Joe's takes product to a new level with a carefully managed product line of approximately 4,000 SKUs [vs. the more traditional and overwhelming 50,000 SKUs]

Other reasons: Compromise Effects

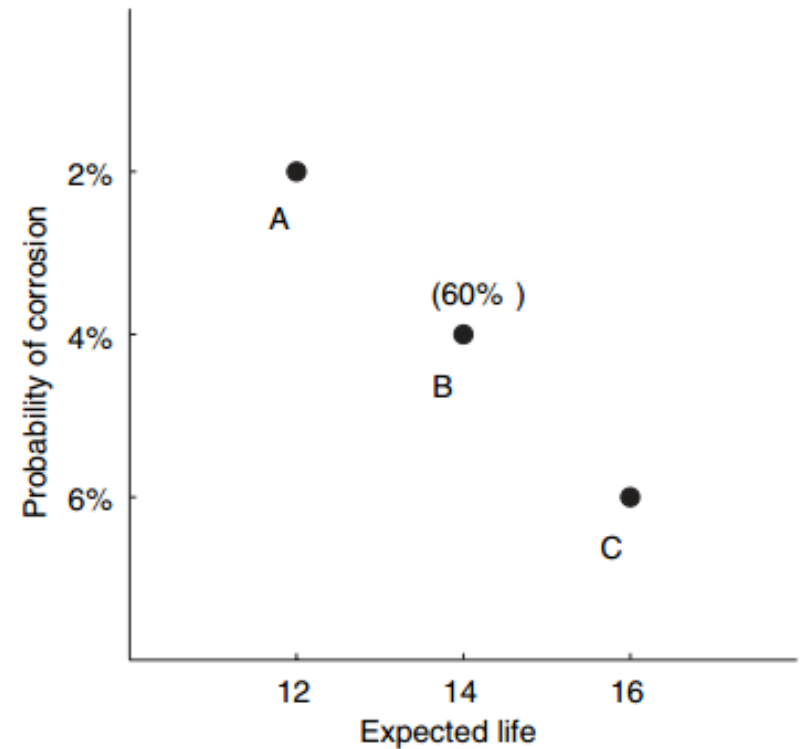
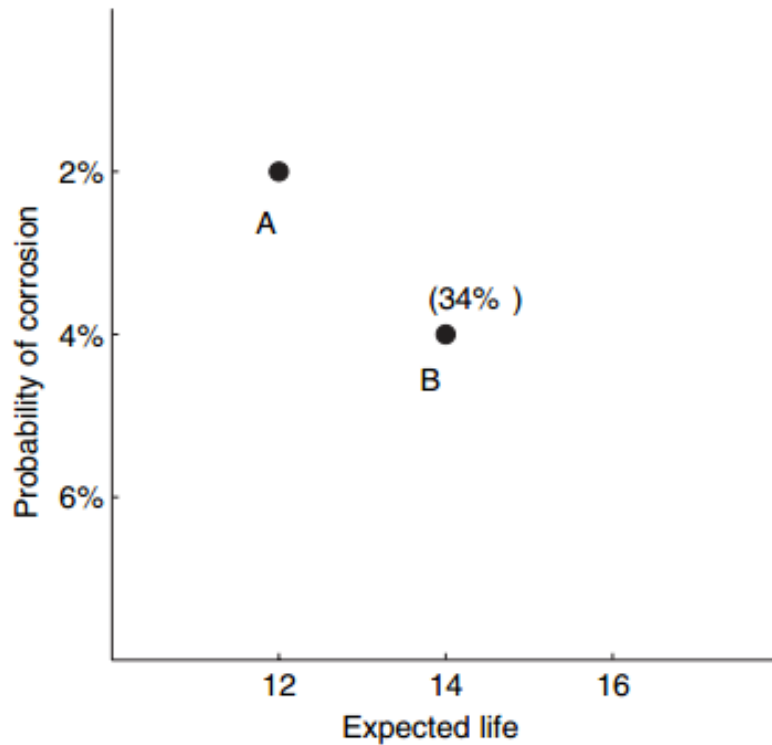


FIGURE 1. COMPROMISE EFFECT WITH CALCULATOR BATTERIES

B sales went up, because people bought it knowing that they are not just going for the cheapest (or most inferior product).

Other reasons: Compromise Effects

Wine Pairing

Offered wine pairing options for your experience. You may also select to pair once seated. All pairings prices will be charged on site at the conclusion of your meal.

Select items

Smyth Traditional Pairing

Smyth's wine pairings take you on a journey that is inspired by classic wine regions while highlighting emerging producers. \$145.00 per person exclusive of...

Smyth Reserve Pairing

Head Sommelier Kevin Goldsmith featuring limited bottlings of elevated quality vineyard sites from deep in the cellar. \$215.00 per person exclusive of tax...

Smyth Super Mega Pairing

Head Sommelier Kevin Goldsmith showcasing a highly rotating selection of allocated wines, exquisite vintages, and often one-off bottles. \$355.00 per person...

Related Products

- Summary
 - Product lines are used to segment consumers
 - Quality is an added dimension
 - Over and above price
 - Tradeoff:
 - Cannibalize your existing product
 - Bring new consumers into the market (either not currently purchasing or from competitors)
 - Behavioral Effects
 - Context effects
 - Choice overload

Summary – New Products and Product Lines

- New Products
 - How do you come up with them?
 - How do you forecast demand?
- Product Line Decisions
 - Sell to new segments vs. Cannibalization
 - It's all just a constrained optimization problem
 - Behavioral components

Summing up Marketing Management

**Customers;
help them meet needs.**

Preferences for product attributes
- over and above the NBA (example, liver transplant)

$U(ij) = d(i)P(j) + X(j)\text{Beta}(i)$
Segment by WTP.

PRICE

**** do not use fixed cost**

**** Q(P), quantity is always a function of P**
EVC, econ val to customers.

**** Always exclude some customers (when we raise price a bit, we lose some. Keep doing till MR=MC).**

Customer does not care about our cost.

Advertising.
Means to achieve other goals.
must use Good Variation.
Be careful about theories —> targeting
- No rule that you must advertise.
- Analyse cannibalisation vs new customers.

Next Best Alternatives.

**Exam; first part is T/F like the final quiz.
second part looks more like first 2 assignments (read PDF then enter canvass quiz, solve offline first).**

Open Q&A

Salesforce compensation.

- the more someone sells, the more you have to incentivise them on the margin to sell the next one. (think of exponential upward graph)

interesting; one of the best ROI is from direct mail (for some industry, one is HR).